Practice Session 01: Cytoscape Basics and Advanced

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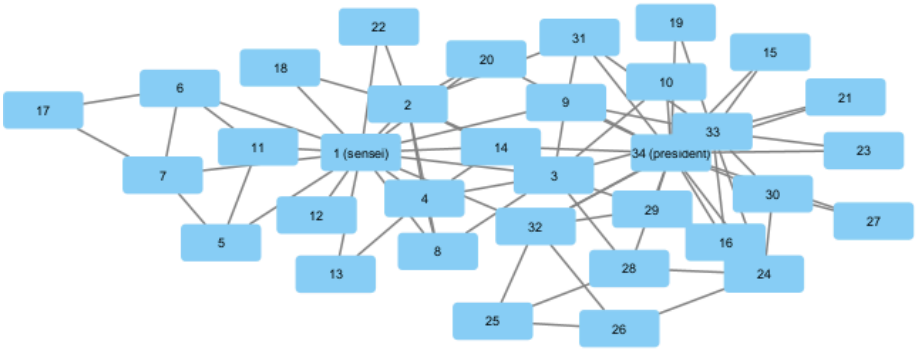
GRAPHS	NODES	EDGES
Zachary's karate club graph	34	78
Reptilia-tortoise graph	787	1713
Second largest connected component	19	130
Third largest connected component	9	49
Line graph	5	4
Marvel Hero graph	7129	167112
Requiem subgraph	8	29
Game of Thrones	84	216
Clique subgraph	6	15
Multi-edge subgraph Lannister	2	2
Multi-edge subgraph Arryn/Baelish	2	3
Philosophers graph	21	27

1. Importing a network

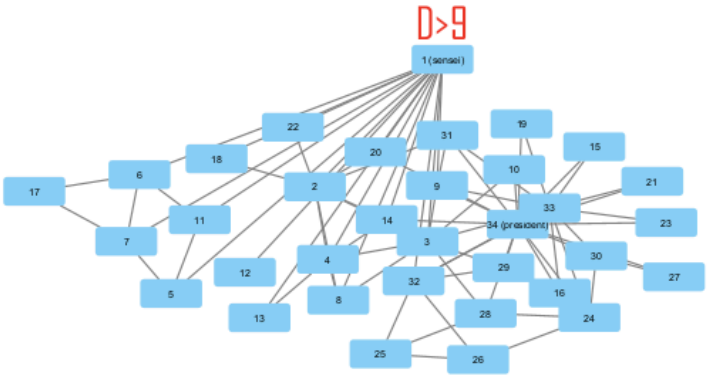
1.1. Import Zachary's karate club

Include in your report this graph plus and a brief paragraph indicating what are the top two nodes having the larger *Degree*. Why are they so connected?

This is Zachary's karate club graph, it has 34 nodes and 78 edges. The two top nodes having the larger degree are President with a degree of 17 and Sensei with a degree of 16. They are so connected because they are the two most important people in the karate class.



Indicate one person that has degree larger than *D=9* .Put the text in red so that we can see.

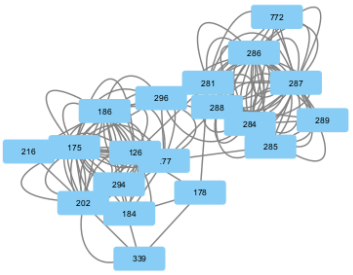


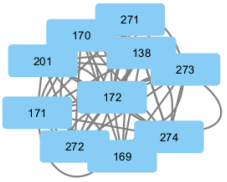
The Compound Spring Embedder is an algorithm derived from force-directed graph layout algorithms. Read the Wikipedia page on [force-directed graph drawing](https://en.wikipedia.org/wiki/Force-directed_graph_drawing) and explain in one paragraph, in your own words, how this works.

Force-directed graph drawing is an algorithm that works by treating the nodes as if they're pushing each other apart, while edges act like springs pulling connected nodes together. The algorithm keeps adjusting the positions until everything balances out. In the end, you get a layout where related nodes are grouped closer, making the graph easier to read and understand.

1.2. Import network of tortoises

Extract the *second and third largest connected components* of this graph, breaking ties arbitrarily if there are components with the same size, and include this in your report.



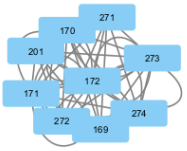


Second largest connected component

Third largest connected component

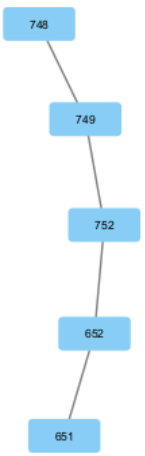
Extract the largest connected component that is a *clique*, include it in your report and indicate how many nodes it has.

The largest connected component that is a clique has 9 nodes and 49 edges.



Extract the largest connected component that is a *line graph*, include it in your report and indicate how many nodes it has.

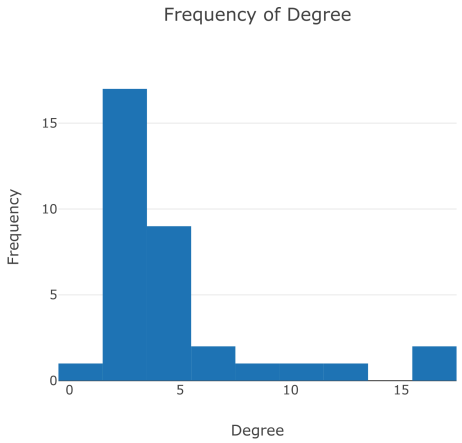
The largest line graph has 5 nodes and 4 edges.



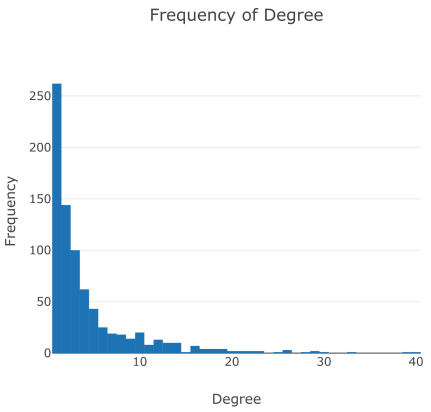
3. Basic network analysis

3.1. Plot degree distributions

Include a plot with the degree distributions in all two graphs (Karate, Tortoises).



Karate

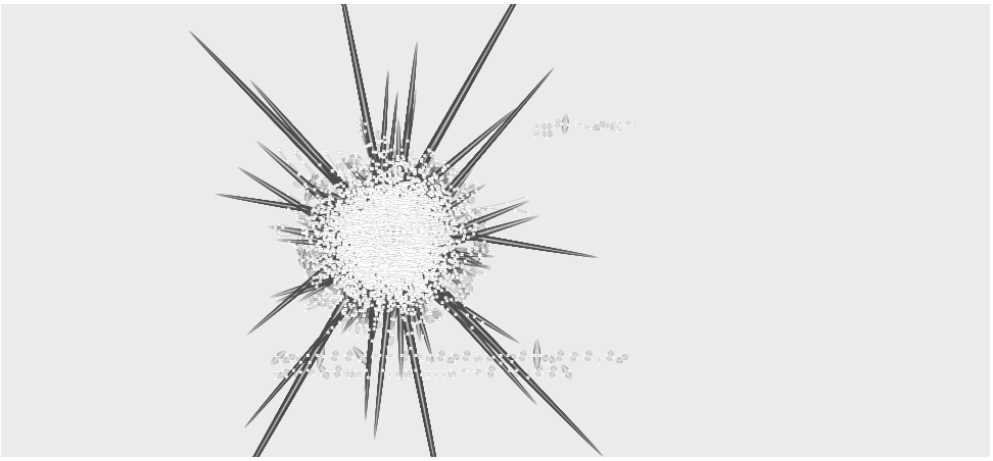


Tortoise

4. Working with a large network

4.1. Import this network into Cytoscape

Include this graph in your report.



What are the top 20 nodes with the largest degree?

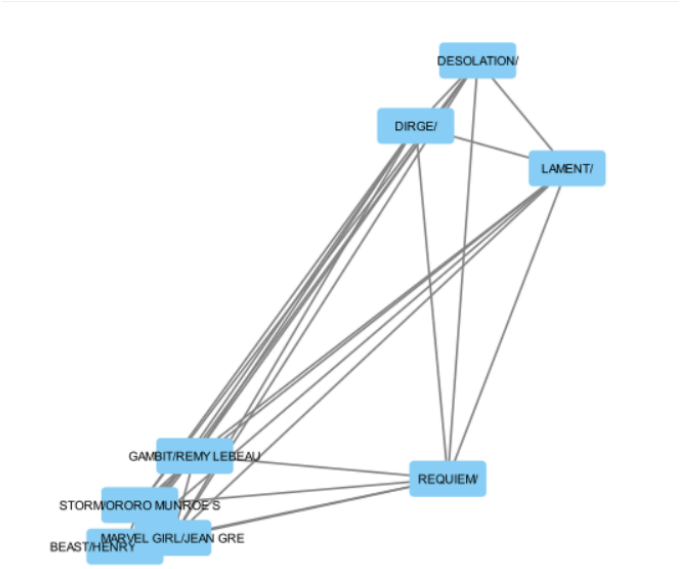
Character	Degree
CAPTAIN AMERICA	1603
JONES	1534

SPIDER-MAN/PETER PARKER	1408
IRON MAN/TONY STARK	1302
THING/BEN GRIMM	1222
WOLVERINE/LOGAN	1203
JAMESON	1186
SCARLET WITCH/WANDA	1170
HUMAN TORCH/JOHNNY STORM	1165
MR. FANTASTIC/REED RICHARDS	1161
THOR/DR. DONALD BLAKE	1139
BEAST/HENRY & HANK PYM	1118
VISION	1091
INVISIBLE WOMAN/SUE STORM	1072
HAWK	1045
CYCLOPS/SCOTT SUMMERS	965
ANGEL/WARREN KENNETH	961
ANT-MAN/DR. HENRY J.	948
STORM/ORORO MUNROE	945
WASP/JANET VAN DYNE	944

4.2. Create sub-graphs

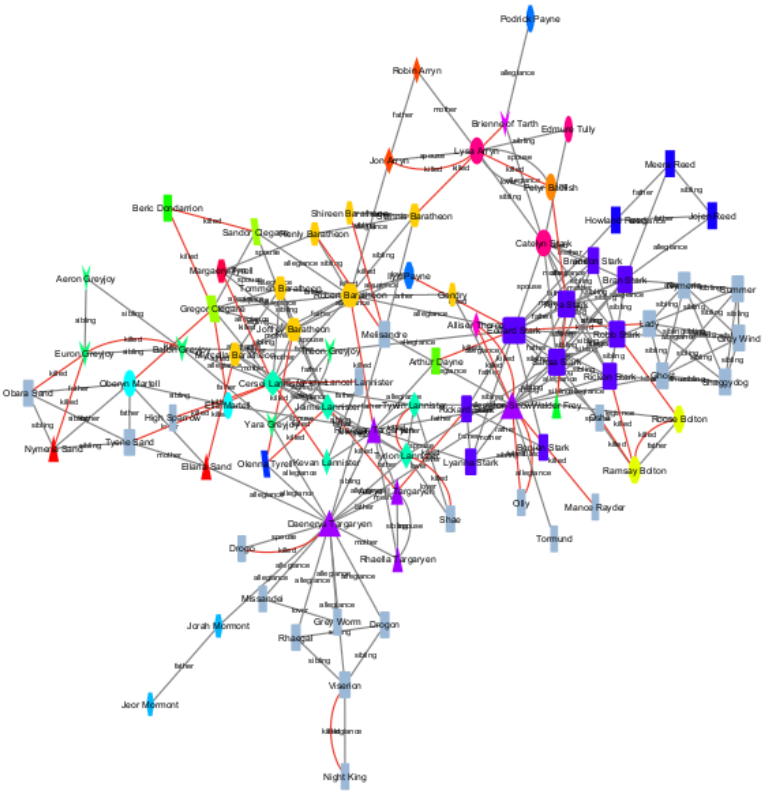
Indicate the number of nodes and edges of this subgraph, it appears on the left panel next to the name of the network.

I chose the Requiem character and the graph has the adjacent characters to it and its connections, it has 8 nodes and 28 edges.

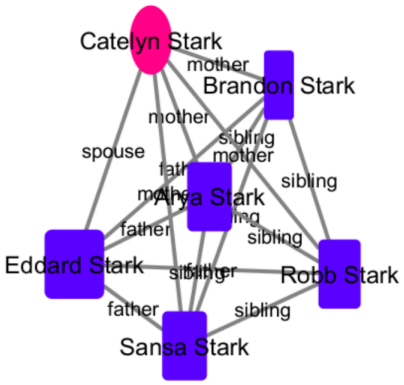


5. Creating a network

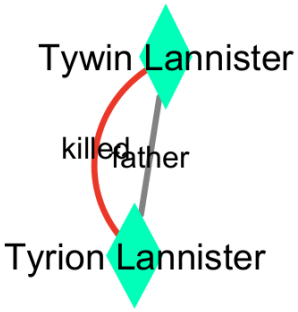
5.1. Creating from existing spreadsheets
Include the graph of *Game of Thrones*.

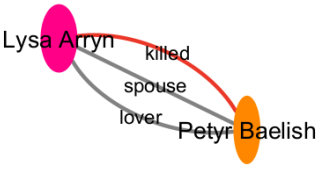


Select and extract a graph containing one clique of four or more characters. Indicate which characters are involved, and style the edges so that they show as label the *relation* between the characters.



Show two examples of a multi-edge in this graph. Indicate which characters are involved. Extract the corresponding (2-nodes) subgraphs.





Describe any observation you can make about how the node attribute "house of birth" is distributed in the graph.
An observation we can make regarding the house of birth is that in most of the cases if they belong to the same house they are usually connected with a family link. Usually people from the same family do not kill each other.
5.2. Create a new network in a CSV file

Describe the graph you created.
The graph I created contains some of the most important philosophers, forming a network of influences. It is clear that most philosophers have been influenced by their predecessors, but in this case, we have only considered the most direct influences, such as Hume's influence on Kant. The attributes of the nodes represent the philosophical schools most scholars agree they belong to, with each school having a different shape and color. I think that a way to improve this graph would be to make it a directed graph to make sure the influences are clear.
Include two tables with the contents of the CSV files you created (not screenshots).

Source	Target
Plato	Aristotle
Plato	Descartes
Plato	Kant

Plato	Hegel
Aristotle	Kant
Aristotle	Aquinas
Descartes	Kant
Descartes	Spinoza
Kant	Hegel
Kant	Nietzsche
Hegel	Marx
Hegel	Sartre
Marx	Engels
Marx	Lenin
Nietzsche	Sartre
Nietzsche	Foucault
Sartre	Camus
Sartre	Heidegger
Wittgenstein	Russell
Wittgenstein	Camus
Socrates	Plato
Socrates	Aristotle
Foucault	Derrida
Russell	Wittgenstein
Camus	Sartre
Hume	Kant
Leibniz	Descartes

Node	Philosophical House
Plato	Idealism
Aristotle	Empiricism
Descartes	Rationalism
Kant	Idealism
Hegel	Idealism
Marx	Materialism
Nietzsche	Existentialism
Sartre	Existentialism
Wittgenstein	Analytic
Aquinas	Scholasticism
Spinoza	Rationalism

Engels	Materialism
Lenin	Materialism
Foucault	Continental
Camus	Existentialism
Heidegger	Continental
Russell	Analytic
Socrates	Idealism
Derrida	Post-Structuralism
Hume	Empiricism
Leibniz	Rationalism

Draw the graph in Cytoscape and include it in your report.



I declare that for some reason in exercise 4 the data I used to complete the exercise is different from the data used by some of my colleagues. I downloaded the file from github but the version they used was different from mine. Other colleagues had the same problem as well.

I hereby declare that all of the text, tables, and figures in this report were produced by myself.